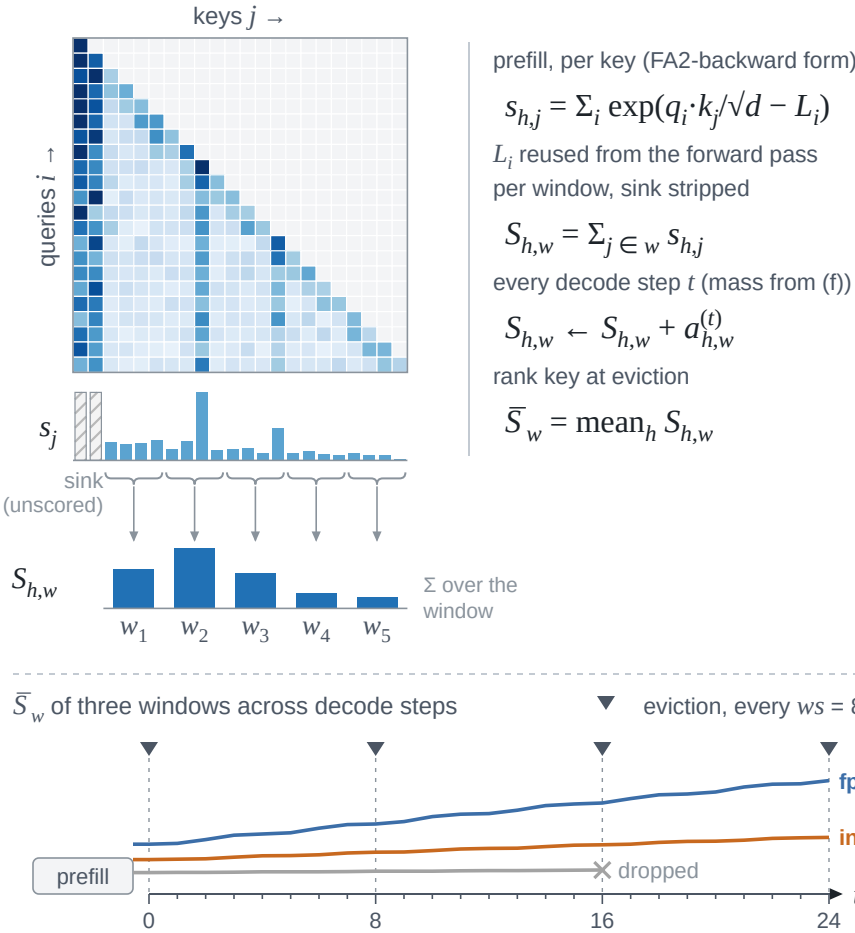


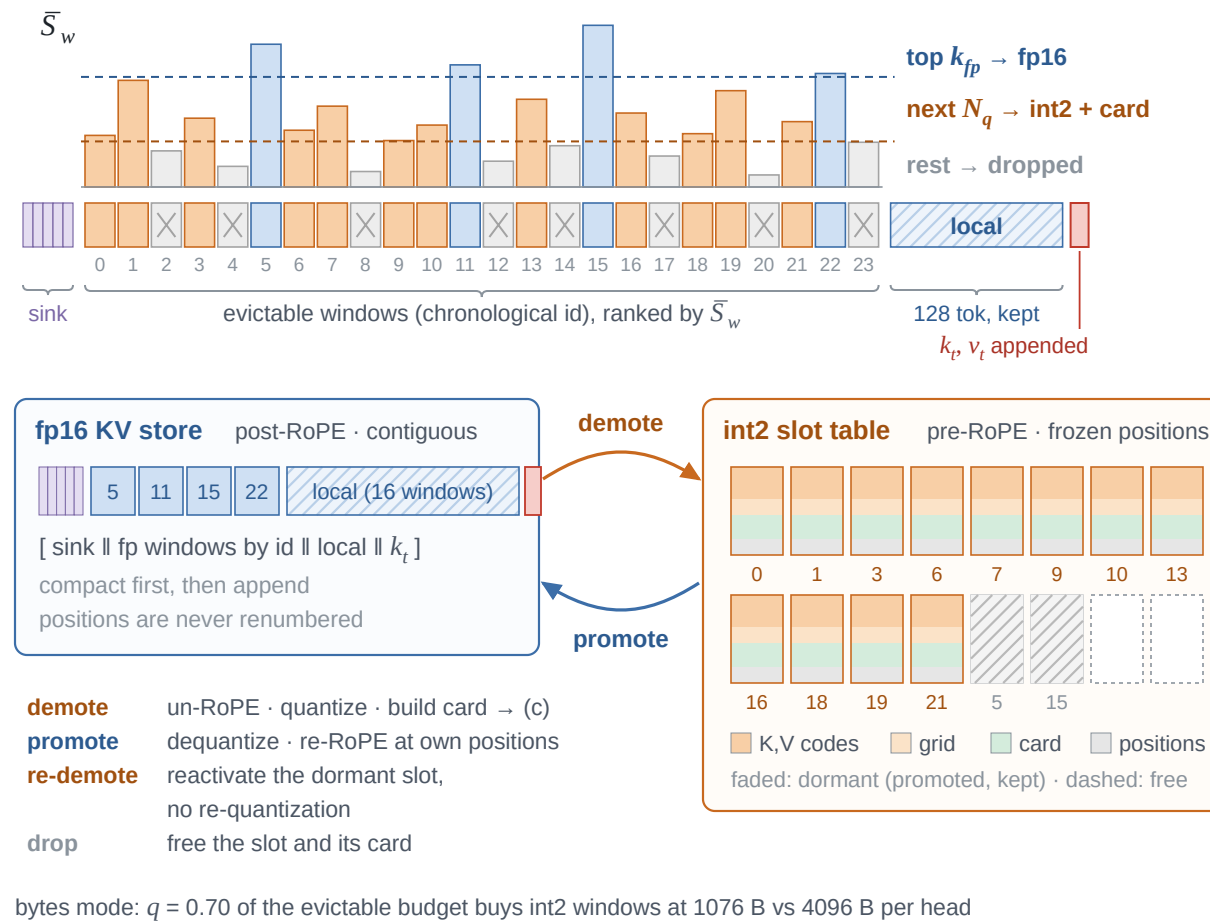
(a) Cumulative window attention

prefill sums attention down the query axis; each decode step adds its mass



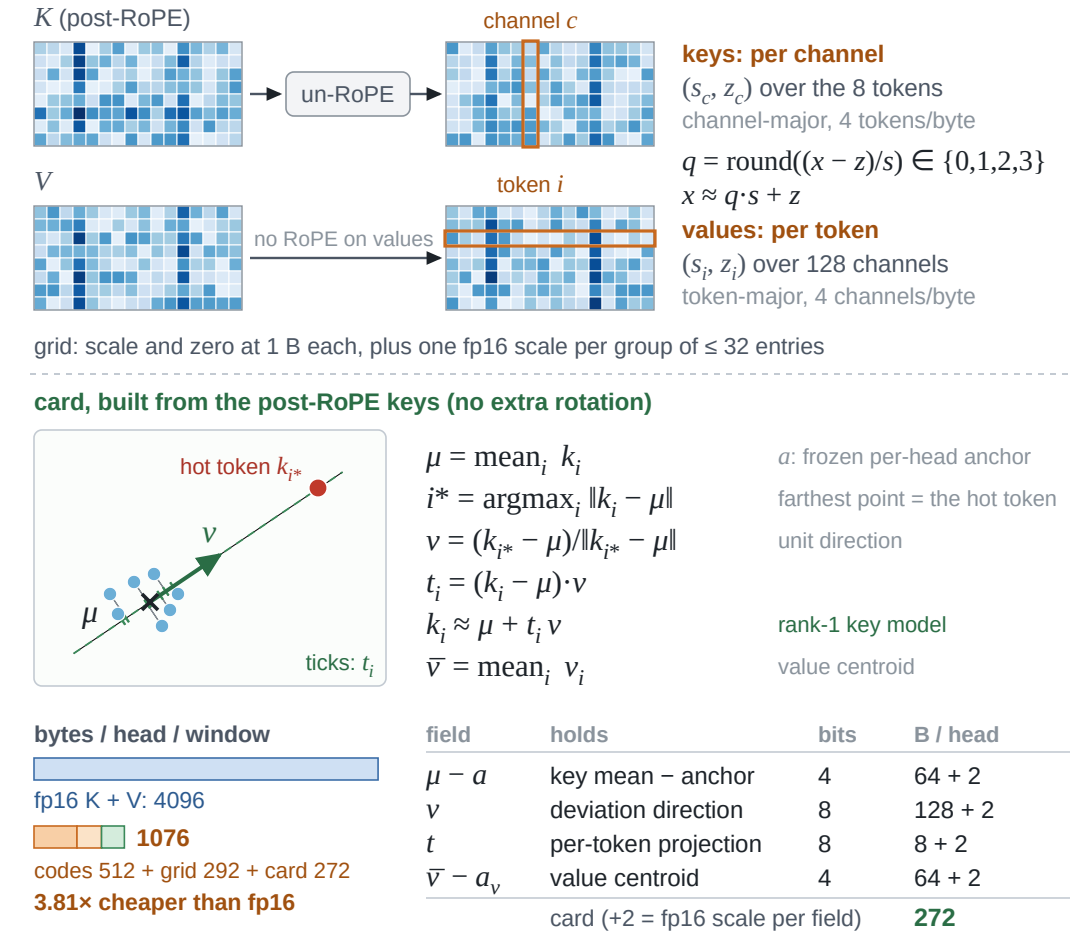
(b) Eviction: rank $\rightarrow$ tier $\rightarrow$ compact

every ws = 8 steps; all L layers fold into $R = L \cdot B$ rows, one compiled pass



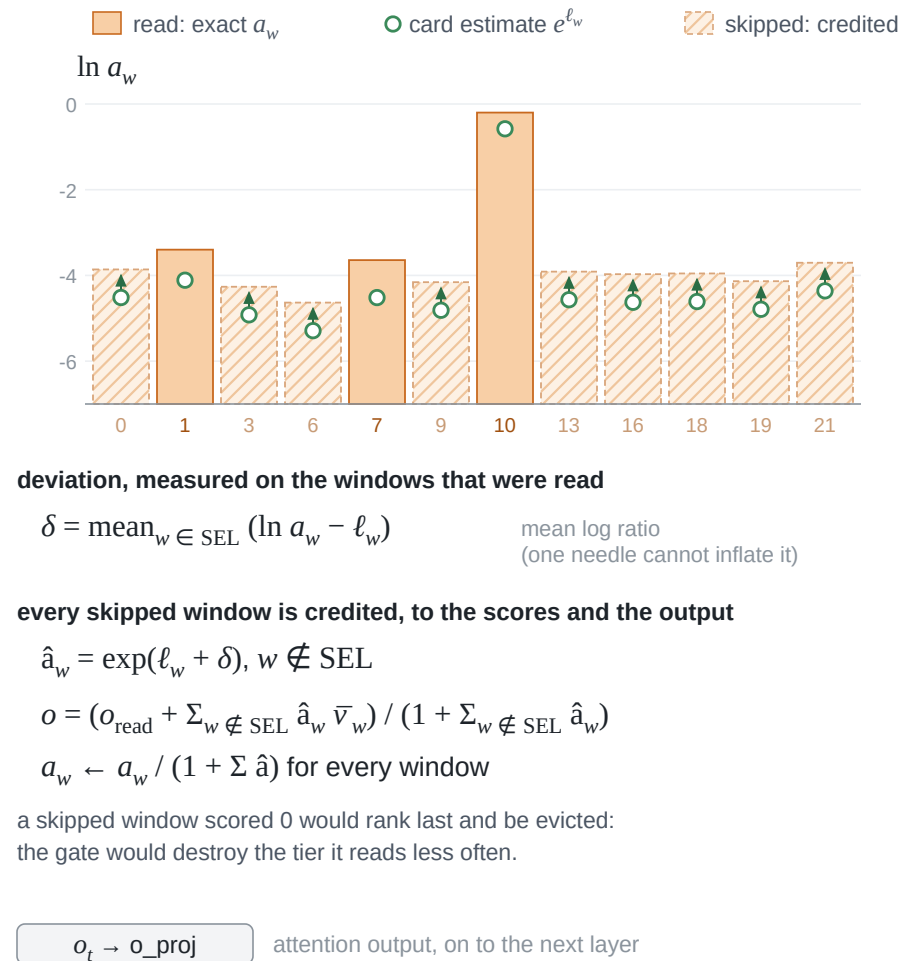
(c) Demotion: 2-bit window + rank-1 card

one window (ws = 8 tokens) of one KV head, $d = 128$ · written once, frozen for life



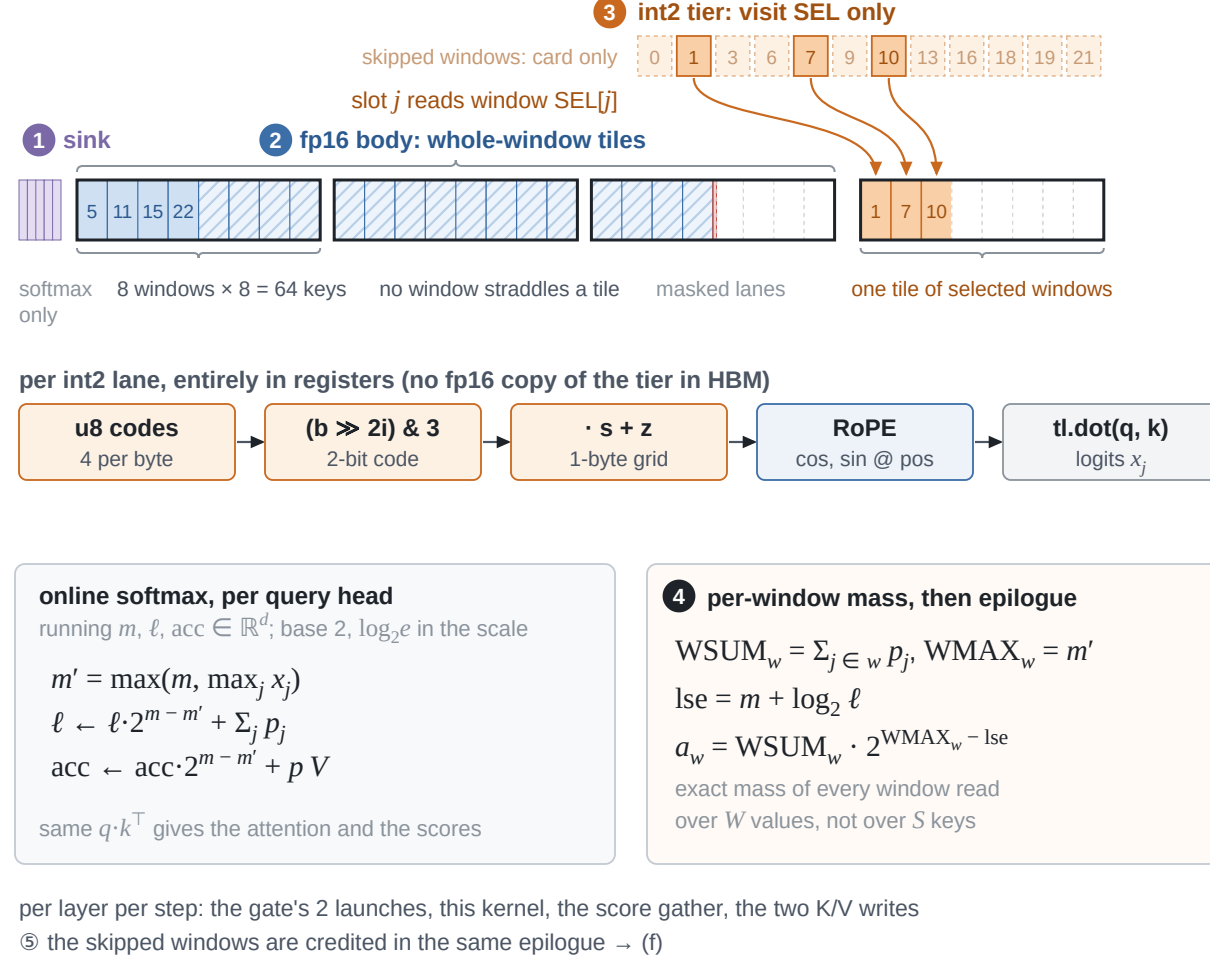
(f) Credit skipped windows, renormalise

kernel epilogue ⑤ · every term is an exact no-op at gate ratio 1.0



(e) Fused two-tier decode kernel

one program per (batch, KV head) serves all 4 query heads: each K/V tile is loaded once



(d) Read gate: score every card, open the top 25%

per KV head: its 4 query heads choose together · 2 launches per layer

